EDUCATION, CURRICULUM AND INSTRUCTION: PH.D. PROGRAM PLANNING FORM
STEM COURSE REQUIREMENTS – EFFECTIVE FALL 2016

Name:	
Track:	
Adviser(s):	

Preliminary Exam Committee: Four faculty members - three (including the student's adviser) representing the major area and one representing the minor field (outside C&I) or supporting program (within or outside C&I).

Names of Major Examiners:		**Names of Minor or Supporting Program Examiners:**

Core Coursework: (6 credits)

Course Designator	Course Title	Term/ Year	Semester Credits
CI 8131*	CI Core: Critical Examination of Curriculum in Context		3
CI 8132*	CI Core: Teaching Theory and Research		3
	Credit Sub-Total		6

STEM Core Coursework: (9 credits)

Course Designator	Course Title	Term/ Year	Semester Credits
CI 8571*	Issues of Equity, Policy, and Social Justice in STEM Education		3
CI 8572*	Learning Theory and Classical Research in STEM Education		3
CI 8573*	Nature of Inquiry in STEM Education		3
	Credit Sub-Total		9

Science Education Focus: (9 credits)

Course Designator	Course Title	Term/ Year	Semester Credits
	Credit Sub-Total		

Mathematics Education Focus: (9 credits)

Course Designator	Course Title	Term/ Year	Semester Credits
	Credit Sub-Total		

Engineering Education Focus: (9 credits)

Course Designator	Course Title	Term/ Year	Semester Credits
	Credit Sub-Total		

Agricultural Education Focus: (9 credits) One additional "AFEE" course at 3 credits is required to fulfill requirements

Course Designator	Course Title	Term/ Year	Semester Credits
AFEE 8090	Seminar: Agricultural Education and Extension		
AFEE 8094	Research in Agricultural Education and Extension		
	Credit Sub-Total		

Research Methodology: (18 credits) CI 8134 and CI 8135 must be taken the first Fall and Spring semester of program.

Course Designator	Course Title	Term/ Year	Semester Credits
CI 8134*	Foundations of Research in Curriculum and Instruction I		3
CI 8135*	Foundations of Research in Curriculum and Instruction II		3
EPSY 8261* OR EPSY 8251**	Statistical Methods I: Probability and Inference OR Methods in Data Analysis for Educational Research I		3
EPSY 8262* OR EPSY 8252**	Statistical Methods II: Regression and the General Linear Model OR Methods in Data Analysis for Educational Research I		3
	Six credits of additional advanced coursework in quantitative and qualitative methods in consultation with adviser		

	Credit Sub-Total		

Minor or Supporting Program: (12-15 credits)

Course Designator	Course Title	Term/ Year	Semester Credits
	Science Education Guidelines 12 credits. No requirements. Minor or supporting field is done on a case-by-case basis in consultation with advisor. **Engineering Education Guidelines** 12 credits. No requirements. Minor or supporting field is done on a case-by-case basis in consultation with advisor. **Mathematics Education Focus** Mathematics minor is determined by the Mathematics Department DGS and is usually 12-15 credits with one 5000 or 8000 sequence. Supporting field in mathematics requires 12 credits at 5000 level determined in consultation with advisor. Other options for supporting field sin mathematics education should be discussed with your adviser.		
	Credit Sub-Total		
	Total of Course Credits (54 minimum)		

* Required
** The 51/52 series is more rigorous and recommended for those considering coursework in HLM etc.

Thesis Credits: 24 Thesis Credits are required (CI 8888). Thesis credits are not graded. Students may start registering for CI 8888 after they have completed at least 36 course credits. Permission for early thesis credit registration is granted by the Graduate Studies Coordinator (cigs@umn.edu) after the student has filled out and submitted the PhD Early Thesis Credit Registration Request.

Once a student registers for their last 54 credits of coursework – and this form is completely filled in – please have the faculty adviser(s) sign the form, then return the signed form to the Graduate Studies Coordinator (125 Peik Hall or cigs@umn.edu). Please note: transfer coursework should be listed on this form as well – please note next to the course what institution the course is being transferred from.

Signature of Adviser(s): _____

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